

What metadata is needed for semantic and data mappings?

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Abstract— A huge amount of data is produced each day through social networks like Facebook, Twitter, etc. Through semantic web technologies, data produced from these social networks can be shared, integrated, and used to gain knowledge. As social networks grow more heterogeneous, declarative data mapping can help address this issue. The use of declarative data mappings is a valuable method of describing the relationship between two different datasets. These mappings are created within the community, but without standard metadata such that quality mappings cannot be shared, discovered, and reused properly. The ultimate purpose of our research is to identify a set of metadata fields/model that can be used to annotate declarative semantic and data mappings to facilitate their discovery, reusability, and quality assessment. As an early step in our research, 18 participants completed an online questionnaire between June and September 2022. The questionnaire targeted participants with varying levels of experience with semantic and data mappings. The survey results indicate that the model we have proposed for mapping metadata has received a positive response from the expert and intermediate-level participants. Also, expert participants have suggested adding extra metadata fields which are being considered as the second version of the model that is being developed.

Keywords— Mapping, Alignment, Metadata Model, Knowledge Representation, Annotation.

I. INTRODUCTION

The World Wide Web (WWW) is widely used as a means of exchanging information. Interoperability between different information systems on the web is limited due to semantic heterogeneity and heterogeneous data. This issue can be observed when working with different datasets and ontologies, as well as when publishing data to the web. As a solution to this issue, declarative mappings have been suggested as a useful method of specifying how two semantics or data models relate to one another [1]. Within linked data and semantic web domains, declarative mapping processes can be classified into three categories, namely ontology mapping, interlinking, and uplift/downlift. The uplift mapping process involves converting different data formats to the Resource Description Framework

(RDF), while the downlift mapping process refers to the reverse [2]. Interlinking involves linking named resources (identified by URIs) in order to indicate how they are related, including if they describe the same concept [3]. Ontology mapping enables the discovery of semantically similar correspondences between different ontologies [1]. These three mapping processes can be found in social networks, for example, when linking social network data sets, mapping similar ontologies, and elevating social network data into RDF. All three mapping processes result in mappings expressed in a declarative manner, although the representation of the declarative mappings may differ. The process of creating these mappings is not trivial since they are intended for a specific purpose and demand expertise, resources, and time [4]. To facilitate their development and reduce the associated cost, the re-use of previously developed declarative mappings is considered necessary. It is argued that identifying appropriate mappings for reuse can be accomplished by examining the metadata associated with these mappings.

Metadata is often referred to as data about data. A clear acknowledgment of the importance of metadata to describe a mapping is found in [5], where Shvaiko emphasizes the need to carefully consider alignment metadata and annotations to provide standard ways to communicate and retrieve alignments. There have been several studies in the literature that have introduced different metadata models and ways to enrich mappings with metadata [6][7][8][4]. Most of these studies provide metadata that focuses only on the final generated mapping. However, a few studies, including [4] and [8], have taken into account the entire lifecycle involved in developing a mapping, not just its final outcome. Furthermore, previous studies have developed metadata models for specific types of mappings, and to the best of the authors' knowledge no previous work have considered a unified metadata model for all type of mappings. Moreover, most existing declarative mappings tend to have no metadata or if they do, have minimal metadata represented as plain text (making them difficult to query and process). These limitations make it challenging for academia and industry to discover and reuse mappings effectively and efficiently.

The ultimate purpose of our research is to identify a set of metadata fields/model that can be used to annotate declarative semantic and data mappings to facilitate their discovery, reusability, and quality assessment. This paper describes the results of an initial survey we have undertaken to evaluate our first proposed metadata model, as described in [9]. In section II an overview of the survey methodology is presented, as well as a brief overview of our proposed metadata model that participants were asked to provide feedback upon. The survey results are presented in section III, followed by section IV, which discusses the results and highlights the main findings. The final section summarizes the results and discusses the future directions of this study.

II. METHOD

In our research, we are attempting to create an approach that will enable the annotation of mappings with a unified machine-readable metadata model that captures the development lifecycle of declarative mappings, irrespective of type. The purpose of the survey described in this paper was to get feedback on our first proposed model [9] with semantic web and linked data community members by examining how relevant the proposed set of metadata fields is for two mapping-related tasks, namely i) finding mappings that can be reused, and ii) evaluating mapping quality. These metadata fields (see Table 1) were identified to be associated with different phases of a proposed declarative mapping lifecycle. A mapping lifecycle breaks down the mapping process into stages that describe the construction of the mappings [9]. six phases: Analysis, Mapping Design, Mapping Development, Mapping Testing, and Mapping Maintenance.

A. Participants

Participation invitations were sent to researchers in the declarative mapping area and through community mailing lists, such as semantic-web@w3.org, the former semantic web interest group. A total of 18 participants agreed to participate in the study so far in the period July 2022 to September 2022. As one of the objectives of this study was to examine whether prior experience influences a person's assessment of the relevancy of metadata fields to mapping-related tasks, participants were asked to describe their level of experience regarding declarative semantic and data mappings. The first question of the survey asked how would you describe your current level of experience in data and semantic mapping. The participants had a choice between no experience, beginners, intermediates, and experts. **Nine described themselves as experts** in semantic and data mapping, **six as intermediates**, and **three as beginners**. However please note that when we analyzed the ratings provided by the 'beginners' cohort, no obvious tendency or common points of agreement could be detected in their ratings of the fields for either task, so we do not present these results as tables in section III, but rather just provide commentary on their results in section IV.

B. Procedure

The survey feedback was collected using a google form that was sent to participants via mailing lists. At the beginning of the survey as explained previously participants were asked to describe their current level of data and semantic mapping experience. Then they were asked to rate the relevance of

various metadata fields to the two mapping-related tasks which were presented as: finding mappings that can be reused, and evaluating quality of a declarative mapping. These tasks were selected based on their importance in a variety of other mapping-related activities, such as supporting the reuse of high-quality mappings, creating new mappings, and maintaining them.

Each task was represented in separate section of the form. The first section of the questionnaire asked participants to rate the relevance of metadata fields for the mapping discovery process, based on their experience. Our provided definition stated that *The metadata fields relevant to the discovery process are expected to facilitate the discovery process when a query is made*. Participants were asked to rate (in their opinion) how relevant the proposed metadata fields are to the mapping discovery process on a scale of 1 to 5. The proposed metadata fields in this section are grouped into 7 categories, each of which corresponds to a phase of the mapping process (see Table 1). The second section of the questionnaire asked participants to rate the relevance of metadata fields for the quality assessment process, based on their experience. Our provided definition stated that *the metadata fields relevant to the quality assessment process are expected to facilitate the quality assessment process when a query is made*. Participants were asked to rate (in their opinion) how relevant the proposed metadata fields are to the mapping quality assessment process on a scale of 1 to 5. The proposed metadata fields in this section are grouped into 7 parts as well , each of which corresponds to a phase of the mapping process. Each section concluded with a question asking participants whether they could suggest any other metadata fields to improve the task.

TABLE 1. PROPOSED METADATA FIELDS

MAPPING LIFECYCLE PHASES	PROPOSED METADATA FIELDS
Analysis	Metadata about stakeholders (URI, Name, Background, Role, organization)
	Metadata about the purpose of the mapping (Requirements, Type of mapping, Mapping domain, Domain assumptions, technical requirement, Risks or issues)
	Metadata to describe inputs that will be mapped (URI, Name, Source, Type, Creator, Format).
Design	Final design decisions, Design decision justification, Quality metrics to consider during the development
Development	(URI, Name, Start/End date of the development, Tools, Mapping method, Mapping algorithm, Format)
Testing	(URI, Name, Testing type, Date/Time, Testing result)
Maintenance	(Publisher name, Publisher source, Version number, Version date/time).

III. RESULTS

The following is an analysis of the questionnaire responses received so far.

A. Task: support finding mapping that can be reused

The following tables summarize the number of fields that were rated as relevant or not for the first task by each category of participants. As mentioned previously, participants were asked to rate (in their opinion) the relevance of proposed metadata fields to the mapping discovery process.

a) Ratings of proposed Metadata fields by Experts

In table 2 and 3 below, the number in the cell indicates the number of people who provided the rating shown in the column header for the metadata field whose name is given at the start of the row in question. So, for example, Table 2 shows that four experts rated the proposed field *Name* for describing stakeholders as relevant (taking together ratings 1 and 2 together) to the task. Further, it was rated neutral (rating 3) by two experts, less relevant by one expert, and irrelevant by two experts. As shown in Table 2, six fields have been proposed to describe stakeholders, namely URI, name, background, role, and organization.

TABLE 2. RESULTS FOR THE MAPPING DISCOVERY TASK -EXPERTS

Mapping discovery task - Experts						
	Proposed fields	1 Highly relevant	2	3	4	5 irrelevant
Metadata fields describing stakeholders	Name	2	2	2	1	2
	URI	3	1	1	3	1
	Role	2	3	3	1	0
	Background	1	1	3	4	0
	organization	2	1	5	1	0
Metadata fields describing the mapping project's purpose	Requirement	5	3	0	0	0
	Type of mapping	6	3	0	0	0
	Domain	4	3	2	0	0
	assumptions	2	4	3	0	0
	Technical Requirement	2	3	4	0	0
	Risks/issues	1	4	2	2	0
Metadata fields describing the input to be mapped	Name	1	4	2	1	1
	URI	5	1	1	1	1
	Source	5	1	2	1	0
	Type	7	0	2	0	0
	Creator	1	3	1	3	1
	Format	4	1	3	0	1
Metadata fields describing the mapping design process	Design decisions	4	2	2	0	1
	Justifications	3	3	2	1	0
	Quality Metrics	4	3	2	0	0
Metadata fields describing the mapping development process	Mapping Name	2	3	2	0	2
	Mapping URI	6	0	2	0	1
	Start date and time	1	3	2	1	2
	End date and time	1	4	2	1	1
	Used tools	3	1	4	0	1
	Mapping method	5	3	1	0	0
	Mapping algorithm	3	3	2	1	0
	Mapping format	6	2	1	0	0
Metadata fields describing the testing process of a mapping	Testing Identifier	3	0	3	1	2
	Testing Name	1	2	4	2	0
	Testing type	3	3	3	0	0
	Testing data and time	1	2	3	2	1
	Testing result	4	3	2	0	0
Metadata fields describing the the management and maintenance of mappings	Publisher name	4	3	1	0	1
	Publisher source	5	0	3	0	1
	Version number	5	2	0	1	1
	Version date/time	5	3	0	0	1

b) Ratings of proposed Metadata fields by Intermediates

TABLE 3. RESULTS FOR THE MAPPING DISCOVERY -INTERMEDIATE

Mapping discovery task – intermediates						
	Proposed fields	1 Highly relevant	2	3	4	5 irrelevant
Metadata fields describing stakeholders	Name	2	3	0	1	0
	URI	2	2	1	1	0
	Role	1	3	0	2	0
	Background	1	3	0	2	0
	organization	1	2	2	1	0
Metadata fields describing the mapping project's purpose	Requirement	4	1	0	1	0
	Type of mapping	4	1	1	0	0
	Domain	5	1	0	0	0
	assumptions	2	3	0	1	0
	Technical Requirement	2	2	0	2	0
	Risks/issues	2	2	0	2	0
Metadata fields describing the input to be mapped	Name	3	3	0	0	0
	URI	4	2	0	0	0
	Source	4	1	1	0	0
	Type	3	1	2	0	0
	Creator	1	2	2	1	0
	Format	1	2	3	0	0
Metadata fields describing the mapping design process	Design decisions	0	3	3	0	0
	Justifications	0	3	3	0	0
	Quality Metrics	0	4	2	0	0
Metadata fields describing the mapping development process	Mapping Name	2	3	0	0	1
	Mapping URI	2	3	0	0	1
	Start date and time	1	0	2	2	1
	End date and time	2	0	1	2	1
	Used tools	1	3	0	2	0
	Mapping method	1	4	1	0	0
	Mapping algorithm	1	2	3	0	0
	Mapping format	1	1	4	0	0
Metadata fields describing the testing process of a mapping	Testing Identifier	2	2	0	2	0
	Testing Name	1	1	2	2	0
	Testing type	3	2	0	1	0
	Testing data and time	1	1	1	2	1
	Testing result	3	2	1	0	0
Metadata fields describing the the management and maintenance of mappings	Publisher name	2	2	2	0	0
	Publisher source	3	1	2	0	0
	Version number	1	3	2	0	0
	Version date/time	3	0	3	0	0

c) other metadata fields recommended by participants to improve the mapping discovery process

There have been six recommendations from experts for adding more metadata fields to the model. Two of the participants suggested reviewing the previously proposed Simple Standard for Sharing Ontology Mappings (SSSOM), which is a simple metadata standard for describing semantic mappings [6]. Several participants have suggested additional fields including publisher URI, mapping constructs, number of projects, data amount, and the change/issue management process.

B. Task: support evaluation of the quality of mapping

The following tables summarize the number of fields that were rated as relevant or not for the mapping quality assessment task by each category of participants.

a) Ratings of proposed Metadata fields by Experts

In table 4 and 5 below, the number in the cell indicates the number of people who provided the rating shown in the column header for the metadata field whose name is given at the start of the row in question.

TABLE 4. RESULTS FOR THE MAPPING QUALITY ASSESSMENT TASK - EXPERTS

Mapping quality assessment task - Experts						
	Proposed fields	1 Highly relevant	2	3	4	5 irrelevant
Metadata fields describing stakeholders	Name	1	3	1	2	1
	URI	2	1	2	2	1
	Role	2	2	3	1	0
	Background	1	1	4	2	0
	organization	2	3	2	1	0
Metadata fields describing the mapping project's purpose	Requirement	7	1	0	0	0
	Type of mapping	6	1	0	1	0
	Domain	4	2	2	0	0
	assumptions	3	5	0	0	0
	Technical Requirement	3	3	1	1	0
	Risks/issues	1	4	1	2	0
Metadata fields describing the input to be mapped	Name	0	3	2	1	2
	URI	2	2	1	2	1
	Source	4	1	1	2	0
	Type	5	1	0	2	0
	Creator	1	2	0	4	1
	Format	3	1	1	2	1
Metadata fields describing the mapping design process	Design decisions	2	5	0	0	1
	Justifications	3	3	2	0	0
	Quality Metrics	4	4	0	0	0
Metadata fields describing the mapping development process	Mapping Name	0	3	1	2	2
	Mapping URI	4	1	1	1	1
	Start date and time	1	2	1	2	2
	End date and time	1	3	1	2	1
	Used tools	4	0	2	1	1
	Mapping method	4	4	0	0	0
	Mapping algorithm	3	3	2	0	0
	Mapping format	2	2	1	1	2
Metadata fields describing the testing process of a mapping	Testing Identifier	2	1	3	0	2
	Testing Name	0	3	3	0	2
	Testing type	5	1	2	0	0
	Testing data and time	1	4	1	1	2
	Testing result	5	2	1	0	0
Metadata fields describing the the management and maintenance of mappings	Publisher name	3	1	1	1	2
	Publisher source	2	0	4	1	1
	Version number	3	2	0	2	1
	Version date/time	3	3	0	0	2

b) Ratings of proposed Metadata fields by Intermediates

TABLE 5. RESULTS FOR QUALITY ASSESSMENT TASK INTERMEDIATE

Mapping quality assessment task – intermediates						
	Proposed fields	1 Highly relevant	2	3	4	5 irrelevant
Metadata fields describing stakeholders	Name	1	3	1	1	0
	URI	0	3	2	1	0
	Role	1	1	2	2	0
	Background	1	4	1	0	0
	organization	1	2	0	3	0

Metadata fields describing the mapping project's purpose	Requirement	2	3	0	1	0
	Type of mapping	2	3	0	1	0
	Domain	3	1	0	1	1
	assumptions	3	3	0	0	0
	Technical Requirement	2	3	0	1	0
	Risks/issues	2	2	2	0	0
Metadata fields describing the input to be mapped	Name	2	4	0	0	0
	URI	3	3	0	0	0
	Source	5	0	0	1	0
	Type	3	2	0	1	0
	Creator	2	1	2	1	0
	Format	1	4	0	1	0
Metadata fields describing the mapping design process	Design decisions	4	1	1	0	0
	Justifications	4	1	1	0	0
	Quality Metrics	4	1	1	0	0
Metadata fields describing the mapping development process	Mapping Name	2	2	2	0	0
	Mapping URI	2	2	2	0	0
	Start date and time	1	2	0	2	1
	End date and time	1	2	0	2	1
	Used tools	2	1	2	1	0
	Mapping method	5	0	1	0	0
	Mapping algorithm	3	2	1	0	0
	Mapping format	1	2	2	1	0
Metadata fields describing the testing process of a mapping	Testing Identifier	4	0	2	0	0
	Testing Name	3	0	3	0	0
	Testing type	5	0	1	0	0
	Testing data and time	4	0	1	1	0
	Testing result	5	1	0	0	0
Metadata fields describing the the management and maintenance of mappings	Publisher name	2	3	1	0	0
	Publisher source	3	1	0	2	0
	Version number	1	4	0	1	0
	Version date/time	3	1	0	2	0

c) other metadata fields recommended by participants to improve the quartl assessment process

One of the expert participants proposed extra fields, including mapping cardinality, mapping justification, and mapping confidence from the SSOM model [6]

IV. DISCUSSION

Our research is focused on identifying a set of metadata fields for annotation of declarative semantic and data mappings to facilitate their discovery, reusability, and quality assessment. We undertook a feedback survey based on our initial model with members of the linked data and semantic web communities.

In the first task, which is to find a mapping that can be reused, **experts** (nine participants) have considered **the majority of the fields relevant** to the task but with three fields considered less relevant by four experts, namely *stakeholder URI*, *stakeholder background* and *input creator*. A total of six **intermediate-level** participants rated **all the fields relevant** to the task but with three fields considered less relevant, which are development of *mapping starting date*, *ending date*, and *testing date/time*. The three participants who were **beginners** rated **thirteen fields irrelevant**, including *stakeholder URIs*, *stakeholder backgrounds*, *stakeholder organizations*, *risks identified before development*, *input creators*, *input formats*, *development starts and ends date*, *tools used during development*, *mapping formats*, *testing ids*, *testing names*, *testing types*, *testing data and times*, and *testing results*.

On the second task, assessing the quality of mapping, **experts** (eight participants) considered **all fields are relevant** but three less relevant, which are *input creator*, *mapping human readable name*, and *development of mapping starting date*. **Intermediate** participants, representing six individuals, rated **all fields relevant** to the task but three fields less relevant: *stakeholder organization*, *development of the mapping starting and ending date*. The **beginners**, which were two participants, rated **14 fields as irrelevant**, including *stakeholder's human readable name*, *stakeholder's URI*, *stakeholder's role*, *stakeholder backgrounds*, *stakeholder organizations*, *domain assumptions*, *risks identified before development*, *input creators*, *input formats*, *beginning and ending dates of development*, *tools used to develop*, *mapping formats*, *testing data and times*, and *publisher name*.

From this analysis it is evident that there was an influence of a participant's level of expertise on his/her assessment of how relevant the fields are despite the small number of participants. A good example of this influence is when experts and intermediate-level participants, which constitute most of the group, rated only three fields as irrelevant for each task, while beginners rated about thirteen fields as irrelevant. Moreover, only experts have suggested extra metadata fields to the model, out of all the other participants.

To conclude, the main finding of this survey is that the model we developed received a reasonably positive response from the expert and intermediate-level participants, and they rated most of the proposed metadata fields as relevant to the tasks identified and suggested additional useful metadata fields. However, the different categories of participants did not agree on those fields that were deemed less relevant. To put it differently, there were no common fields among the fields that the experts and intermediate-level participants considered irrelevant, at this stage of development of the metadata model. In regard to the extra metadata fields proposed by the experts for both tasks, two experts have suggested considering the SSOM model [6]. However the SSOM model focuses mainly on the actual mapping, rather than the mapping development lifecycle as a whole. Regardless, what has been suggested in their model can be reused and added to the group of metadata fields we propose to describe the mapping development process. As well as this, other experts have suggested adding the *publisher identifier*, *mapping construct*, *amount of data being mapped*, and *changes during the maintenance process*, which will all be considered when the second version of the model is developed.

V. CONCLUSION

This paper presents the results of a questionnaire survey on metadata model for declarative semantic and data mappings. It was conducted to verify a set of predefined metadata proposed by the authors as part of an attempt to develop a machine-readable metadata model that captures the entire mapping development lifecycle so that annotations of mappings can be generated and queried. In this study, 18 participants completed an online questionnaire between June and September 2022. The

feedback gathering is ongoing. Your participation is appreciated at <https://rb.gy/kfpckr>. The questionnaire targeted participants with varying levels of experience with semantic and data mappings from the linked data and semantic web community members. The initial survey results illustrate that the model we have developed has received a positive response from the expert and intermediate level participants, and they rated most of the proposed metadata fields as relevant to the tasks identified. In addition, expert participants have suggested adding extra metadata fields which will all be considered when the second version of the model is developed.

Next steps we will look at how the metadata will be represented and associated with declarative mappings. As a starting point, we will investigate RDF* and Named graphs as possible ways to represent our metadata model. In the following steps, an annotation tool will be developed, and its usability and accuracy will be evaluated by engineers. Finally, we hope to develop a collaborative mapping platform that allows semantic web and linked data experts to share, update, and evaluate mappings.

ACKNOWLEDGMENT

The first author acknowledges financial support from Al Imam Mohammad Ibn Saud Islamic University and the Saudi government - represented by the Royal Embassy of Saudi Arabia - Cultural Bureau in Dublin. The second author is partially supported by the SFI ADAPT Research Centre (grant number 13/RC/2106_P2).

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